

Reactive processing of polymers: Effect of bifunctional and tri-functional comonomers on melt grafting of glycidyl methacrylate onto polypropylene

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ARTICLE INFO

Article history:

Available online 26 November 2009

Keywords:

Reactive processing
Grafting GMA on PP
Grafting with reactive comonomers

ABSTRACT

Two reactive comonomers, divinyl benzene (DVB) and trimethylolpropane triacrylate (TRIS), were evaluated for their role in effecting the melt free radical grafting reaction of the monomer glycidyl methacrylate (GMA) onto polypropylene (PP). The characteristics of the GMA-grafting systems in the presence and absence of DVB or TRIS were examined and compared in terms of the yield of the grafting reaction and the extent of the main side reactions, namely homopolymerisation of GMA (poly-GMA) and polymer degradation, using different chemical compositions of the reactive systems and processing conditions. In the absence of the comonomers, i.e. in a conventional system, high initiator concentrations of peroxides were typically required to achieve the highest possible GMA grafting levels which were found to be generally low. Concomitantly, both poly-GMA and degradation of the polymer by chain scission takes place with increasing initiator amounts. On the other hand, the presence of a small amount of the comonomers, DVB or Tris, in the GMA-grafting system, was shown to bring about a significant increase in the grafting level paralleled by a large reduction in poly-GMA and PP degradation. In the presence of these highly reactive comonomers, the optimum grafting system requires a much lower concentration of the peroxide initiator and, consequently, would lead to the much lower degree of polymer degradation observed in these systems. The differences in the effects of the presence of DVB and that of TRIS in the grafting systems on the rate of the GMA-grafting and homopolymerisation reactions, and the extent of PP degradation (through melt flow changes), were compared and contrasted with a conventional GMA-grafting system.

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1. Introduction

In-situ melt grafting reactions of polar functional groups on commercial inert hydrocarbon polymers such as polyolefins and their copolymers has been the subject of intense studies over the last three decades with the aim of producing a new range of specialty materials with targeted properties [1]. Polypropylene, PP, benefits from its low cost and versatile set of properties hence its use in many commercial applications. However, it is an inert hydrocarbon polymer that lacks the presence of reactive functionality and this limits its application in many technologically important fields due to, for example, its low surface energy, poor dyeability and compatibility with polar polymers. Various reactive (vinyl-based) modifiers have therefore been employed to functionalise PP to introduce new properties that would expand its uses in new areas of applications such as in compatibilisation of polymer blends, interfacial coupling to polymers and inorganic substrates, crosslinking and reactive stabilisation [2–6].

Most attention has been given to free radical melt grafting on PP of two such vinyl-based monomers, maleic anhydride, MA, and glycidyl methacrylate, GMA. These monomers are bifunctional monomers containing an unsaturated function capable of free radical grafting on hydrocarbon polymers and a highly electrophilic moiety that can react with different nucleophilic functions such as carboxylic, hydroxide and amine groups, and this in turn can then be used to promote further reactions, e.g. reactive compatibilisation in blends and alloys [4]. The grafting efficiency of both GMA and MA on different hydrocarbon polymers, however, has been shown to be generally low [2,3,7–9]. In the case of GMA, this is mainly due to its low reactivity towards macroradicals and high tendency towards homopolymer formation.

In our laboratory, we have developed a highly effective and versatile functionalisation strategy that is capable of improving the grafting efficiency of variety of functional monomers on polymers. This approach is achieved through in-situ co-grafting of a small amount of a more reactive (compared to the modifier) comonomer (co-agent) [10–15]. In earlier work, we have reported the use of the polyfunctional comonomer, trimethylolpropane triacrylate, TRIS, with different reactive modifiers, e.g. with GMA in EPR [13] and with various reactive antioxidants in PP [10–12], and have shown

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that these systems give rise to a dramatic increase in the level of grafting of the modifiers on the polymers examined.

We report here our work on the effect of two reactive comonomers, a bifunctional one, divinyl benzene, DVB, and a tri-functional one, TRIS on the characteristics of free radical melt grafting reactions of GMA on PP and to compare these grafting reactions with similar ones but which are carried out in the absence of a comonomer (conventional system).

2. Experimental

2.1. Materials

Granules of polypropylene Apryl-3020 BNI (PP) with melt flow rate of 2.0 g/min (at 230 °C/2.16 kg) were kindly supplied by Elf-Atochem. GMA (97% purity), DVB and TRIS were used as received from Aldrich Chemical Co. The peroxides used were 2,5-dimethyl-2,5-bis(*t*-butyl peroxy) hexane (Trigonox 101, T101), 1,1-di(*tert*-butylperoxy)-3,3,5-trimethyl cyclohexane (Trigonox T-29B90) were kindly donated by Akzo Nobel, dicumyl peroxide (DCP) and benzoyl peroxide (BPO) were purchased from Aldrich and Fluka, respectively, and were all as received without further purification. Analytical grade chemicals and reagents were used for titration. All other chemicals and solvents were of general purpose grade.

2.2. Reactive processing of PP with GMA in presence and absence of comonomers and melt flow measurement

The grafting reaction of GMA on PP was carried out in an internal mixer (RAPRA-Hampden torque rheometer). The pre-heated mixing chamber was initially flushed with oxygen-free nitrogen before introducing polymer samples containing a mixture of GMA and peroxide (T101, unless indicated otherwise) at different molar ratios (mr) along with DVB or TRIS if used (at a specified weight ratio to GMA). Processing was carried out in a closed mixer at 180 °C, unless indicated otherwise, for 10 min with a rotor speed of 65 rpm. The processed polymer was cooled under a stream of nitrogen and stored in a freezer until required for further analysis. Melt flow index (MFI) measurements were conducted on Davenport melt flow indexer using a weight of 2.16 kg at 230 °C in accordance to ASTM D-1238.

2.3. Purification of the grafted polymer and determination of grafting yield of GMA

To ensure the accuracy of the reported level of GMA-grafting yield, the reaction products and free (unreacted) reactants were separated and removed prior to the determination of grafting yield of GMA. Initially, grafted polymer samples were Soxhlet extracted (24 h) with xylene. When the comonomers (DVB or TRIS) were used, the homopolymer of TRIS (poly-Tris) or DVB (poly-DVB), the copolymer of GMA and DVB or TRIS (DVB-co-GMA or TRIS-co-GMA) were separated out as xylene-insolubles. The xylene-soluble fraction was precipitated in excess acetone to give a soluble fraction containing any unreacted monomer/comonomer (GMA, and DVB or TRIS when used) and GMA-homopolymer (poly-GMA), whereas the insoluble fraction contained the GMA-grafted polymer (PP-g-GMA). The 'purified' acetone-insoluble PP-g-GMA polymer was then dried overnight in a vacuum oven at 80 °C and stored until further analysis. This method was used throughout for determining the yield of graft GMA. To calculate the amount of poly-GMA formed during the grafting reaction, a second separation route was followed. The xylene-soluble fraction was precipitated in this case using excess methanol. The poly-GMA co-precipitates with the GMA-grafted

polymer, and the amount of poly-GMA formed was determined from the difference of the two polymer separation methods.

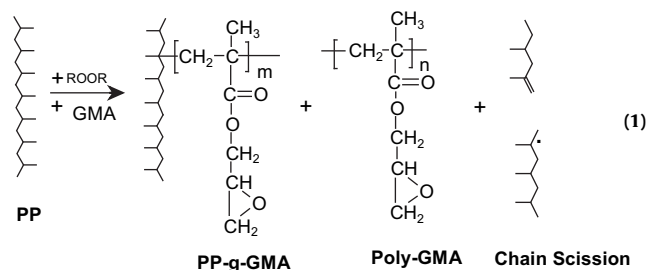
2.4. Determination of the grafting yield of GMA

When GMA was grafted onto the polymer in the absence of the co-agent, compression moulded films of the separated reaction products (i.e. acetone precipitated polymer fractions) were characterised by FTIR and the grafting degree was determined from FTIR measurements using a calibration curve which correlates FTIR data to titration results as described previously [13]. Infrared measurements were performed on a Perkin Elmer Spectrum-1 spectrophotometer over the range 4000 cm^{-1} –400 cm^{-1} with a resolution of 4 cm^{-1} and spectral collection over 64 scans. The carbonyl absorption at 1731 cm^{-1} was used to determine the extent of grafting using the 840 cm^{-1} band (due to $-\text{CH}_2-$ absorption of the PP polymer) as a reference. The relative amount of grafted GMA was calculated from the ratio of absorption areas of peaks at 1731 cm^{-1} and 840 cm^{-1} ($A_{1731 \text{ cm}^{-1}}/A_{840 \text{ cm}^{-1}}$). The level of grafting was determined from the calibration curve of the relative amount obtained by FTIR with the amount of grafted-GMA determined by titration [13]. The degree of grafting (yield) is the mass percent of grafted monomer in the polymer whereas the grafting efficiency is defined as the percent conversion of the total functional monomer initially added to the actual amount of grafted monomer achieved.

3. Results and discussion

3.1. Conventional PP-GMA grafting system in absence of comonomer

In a conventional peroxide-initiated GMA-PP grafting system (i.e. in absence of a comonomer), in addition to the target grafting reaction, there is typically at least two other competing side reactions involving GMA-homopolymer formation (poly-GMA) and degradation of PP mainly by chain scission, reaction (1).



The peroxide (Trigonox 101)-initiated GMA-PP conventional melt grafting system was examined at different peroxide and initial GMA concentrations and at different processing temperatures. The effect of peroxide concentration is shown in Fig. 1. It is clear that at 180 °C increasing the peroxide concentration increased the level of GMA grafting, but this was paralleled by a significant increase in the extent of the main side reaction, poly-GMA, which is formed to a higher level than the amount of grafted GMA (g-GMA) at all peroxide concentrations examined. Higher peroxide concentrations, which are needed in this system to achieve higher grafting levels, also give rise to oxidative degradation of the polymer which is dominated by β -chain scission reactions (via the tertiary macroalkyl radicals, see Reaction (1)) as indicated by the observed lower torque levels (Fig. 1b) associated with lower polymer melt viscosity and is further supported by the observed sharp increase in the melt

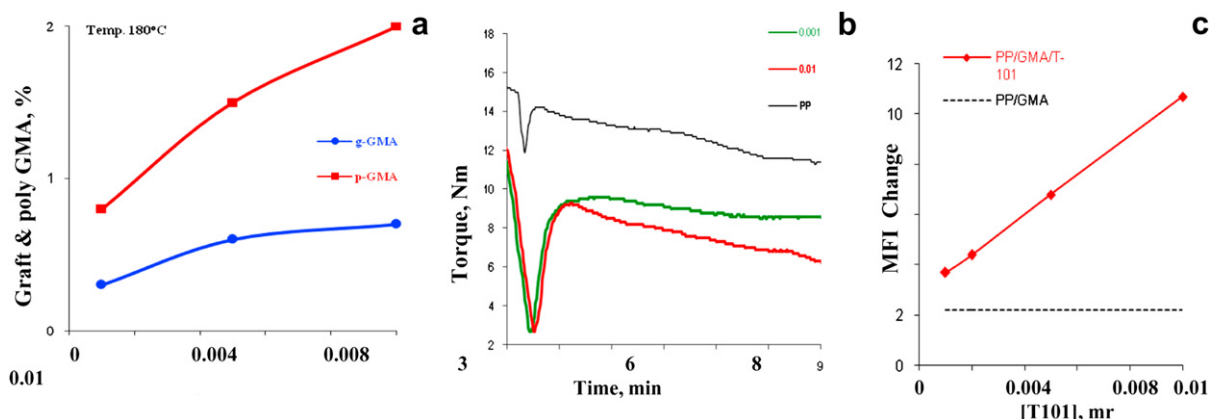


Fig. 1. Effect of peroxide (T101) concentration on characteristics of the conventional GMA grafting reaction (initial [GMA] = 10%, without comonomer).

flow index, Fig. 1c. Increasing the processing temperature has also an adverse effect on the grafting reaction. Fig. 2 shows that an increase in temperature results in a decrease in the g-GMA level which is concomitant with a sharp increase in the concentration of poly-GMA that exceeds the level of grafting. For example at a processing temperature of 200 °C, the grafting level dropped by a factor of 4 and the poly-GMA concentration became 10 times higher than that achieved at the lower temperature of 160 °C. The temperature effect is also associated with increasing extent of PP-chain scission reflected by lowering of the melt flow index (not shown here). Similar results were obtained with increasing the initial GMA concentration in the polymer system where both homopolymer formation and PP-chain scission accompanies the grafting reaction. Fig. 3 a shows that increasing the amount of GMA gives rise to an increase in the level of both g-GMA and poly-GMA with the concentration of the g-GMA being slightly higher, but with further increase in the initial amount of GMA, there is a clear change-over in concentrations so that above an initial GMA concentration of 10%, the poly-GMA level starts to increase rapidly and at 18% GMA its concentration becomes triple that of the grafting level. The sharp increase in the poly-GMA concentration at

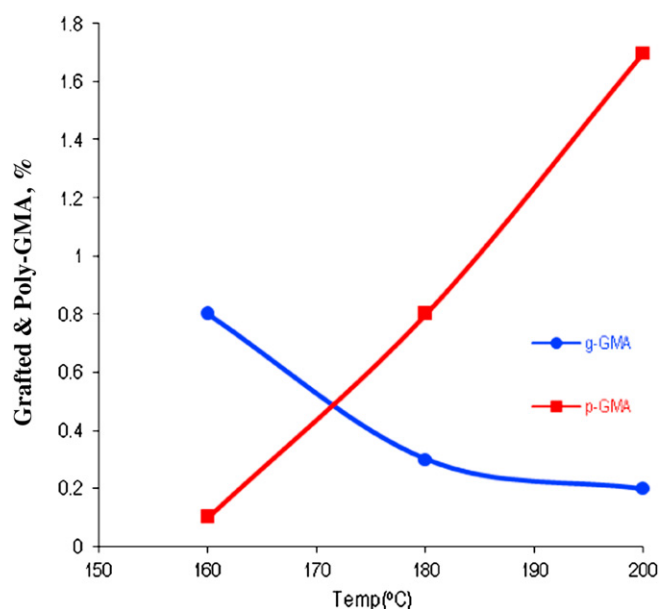


Fig. 2. Effect of processing temperature on extent of grafting and homopolymerisation of GMA (T101 = 0.001 mr, without comonomer).

higher initial amounts of GMA in the system is due to the greater propensity of the GMA–GMA radical coupling reaction at the higher GMA levels which facilitates further the GMA-homopolymer formation at the expense of the target GMA-PP grafting reaction. Examination of the kinetics of the melt reaction (160 °C; initial GMA concentration of 12%; molar ratio of peroxide: GMA of 0.005) carried out by analysing samples at time intervals during processing, Fig. 4, showed clearly that, from the onset of the reaction, homopolymerisation proceeds in direct competition with the grafting reaction. The rate of homopolymerisation at the initial stages of the reaction was much faster than that of the grafting but, with continued processing, the concentration of poly-GMA started to decrease while that of the g-GMA continued to increase but levels off at a concentration lower than that of the poly-GMA.

It is clear from these results that in the PP conventional grafting system both peroxide concentration and processing temperature have an important role in controlling the level of GMA grafting, poly-GMA formation and the extent of polymer degradation. Compared to the effect of peroxide concentration, the processing temperature has a more severe effect on the grafting degree than it is on the extent of polymer chain scission (see Figs. 1 and 2), but it is clear that in both cases the homopolymerisation of GMA competes favourably with the grafting reaction, and only at the lowest processing temperature of 160 °C and lowest peroxide concentration (of 0.001 molar ratio) examined could the grafting degree exceed the poly-GMA level, but, under these conditions the grafting level is very low; a grafting efficiency of only about 6% was achieved. It is important to point out here that under all conditions examined, the overall level of GMA grafting remained relatively low. Low grafting efficiency of GMA in polyolefins and their copolymers has also been reported in the literature and homopolymerisation of different modifiers containing unsaturated reactivity has also been shown in many other systems to be the one most important contributory factor to the low grafting levels of such polymerisable monomers [2,3,13,16].

3.2. PP-GMA-Comonomer grafting system processed in presence of DVB or tris

The above discussion has highlighted the problems of the overall low GMA grafting efficiency on PP, and the need for using high peroxide concentrations required to achieve higher conversion rates that occur at the expense of severe degradation of the polymer and the formation of high levels of poly-GMA. We have shown previously [13] that the free radical grafting yield of GMA on ethylene-propylene rubber is substantially improved by the

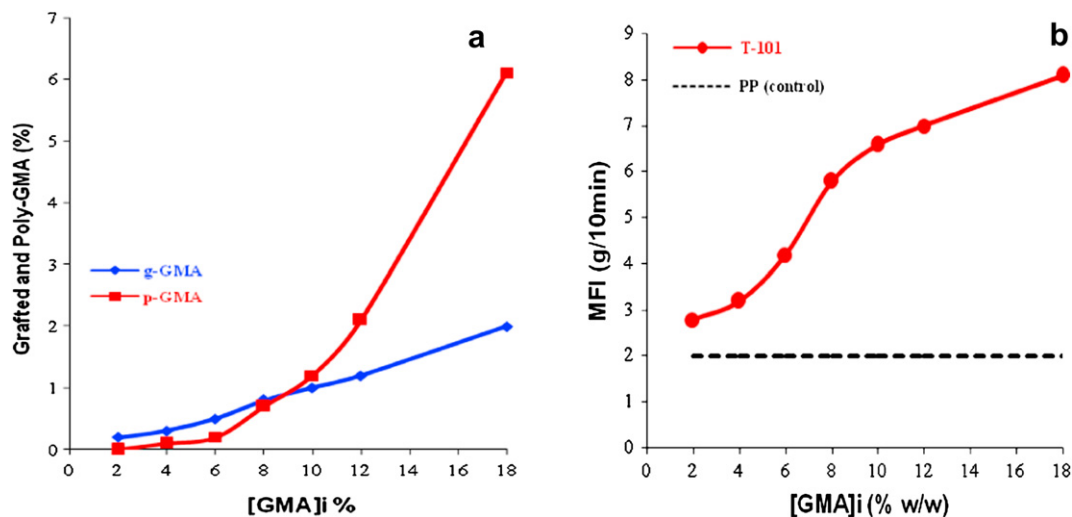


Fig. 3. Effect of initial GMA concentration on extent of grafting and homopolymerisation of GMA (a) and on MFI of the polymer (b); processing temperature = 180 °C, without comonomer.

addition of a small concentration of the highly reactive polyfunctional comonomer trimethylolpropane triacrylate, Tris. Reactive monomers containing more than one polymerisable group have been typically used as crosslinking agents for polymers (due to their high free-radical reactivity) [17,18]. For such highly reactive comonomers containing polymerisable groups to work in assisting the grafting of less reactive functional monomers, the comonomer has to be more reactive towards the polymer macroradicals than the monomer and it should also have a high efficiency towards copolymerisation with the monomer so that it can act as an inter-linking agent between the polymer and monomer resulting in increased grafting efficiency. This would, therefore, require a delicate balance between the composition of the chemical system, e.g. type of initiator, molar ratios of the reagents (monomer, comonomer, initiator) and the process conditions [14,15].

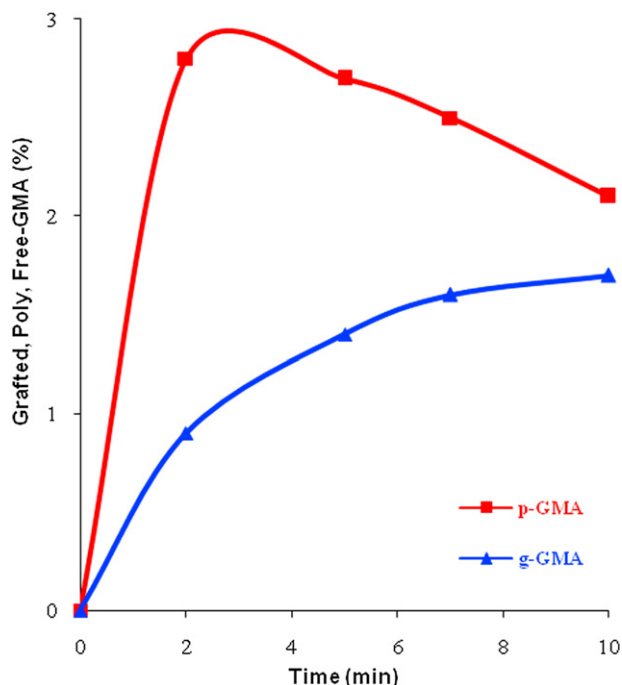


Fig. 4. Kinetics of the melt grafting reaction of GMA ([GMA] = 12%, [T101] = 0.005 mr, processing temperature = 160 °C, without comonomer).

3.2.1. Effect of chemical composition and process variables on PP-GMA grafting reactions in presence of the comonomer DVB

The role of the bifunctional divinyl benzene (DVB) comonomer on the nature and yield of the grafting reaction of GMA on PP under different reaction conditions is examined and compared with the similar PP-GMA conventional (in absence of comonomer) grafting system. The effect of DVB on the torque behaviour of the free radical-initiated melt reactions of PP in the presence and absence of GMA (using small concentration of T101; 0.005 molar ratio, mr, to GMA, and DVB when the latter is used) is compared with that of the conventional PP-GMA grafting reaction, Fig. 5. The reactions containing DVB, in presence and absence of GMA, show distinct high torque peak maxima at the early stages of the melt reactions. At longer processing times, the torque maximum (peak) values in the DVB-reactions drop to much lower values and reach to a level slightly higher than that of the GMA conventional system. The presence of such high torque peak, which is much higher than the

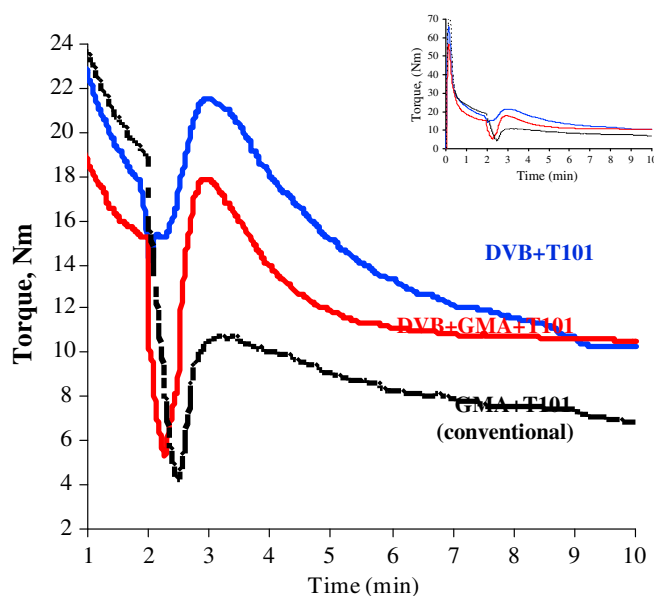


Fig. 5. A comparison of the torque behaviour of GMA-DVB and GMA-grafting systems ([GMA] = 12%, ratio of GMA:DVB = 8:2 w/w, [T101] = 0.005 mr, $T = 180$ °C, without comonomer).

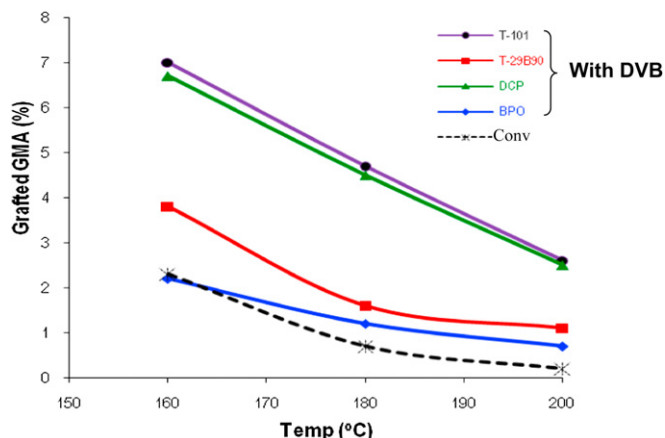


Fig. 6. Effect of processing temperatures on the extent of GMA grafting in the presence of DVB processed with different peroxide initiators ([GMA] = 12%, ratio of GMA: DVB = 8:2 w/w, [T101] = 0.005 mr).

maximum torque of the GMA-PP conventional grafting system (processed with same amount of peroxide), indicates clearly the high free-radical reactivity of DVB towards the PP-macroradicals which is higher than that of GMA-macroradical reaction. The torque peak observed in the DVB/T101 (in absence of GMA) system is attributed to reaction of DVB with the polymer macroradicals resulting in comonomer-assisted branching and crosslinking,

whereas, in the presence of GMA the lower torque maximum indicates the occurrence of both grafting and branching reactions.

The effect of processing temperature in the presence of different peroxides on the level of grafting in the GMA-DVB system is similar to that seen earlier in the conventional system (in absence of DVB, see Fig. 2) in that higher temperatures give rise to lower extent of grafting, Fig. 6. However, there is one important major difference between the DVB and conventional grafting systems in that, in the presence of DVB, the extent of grafting is seen to be much higher, at all temperatures and with all peroxides examined, compared to that achieved in the absence of DVB (in conventional system). It is also clear from Fig. 6 that, compared to the other peroxides, both Trigonox 101 (T101) and dicumyl peroxide (DCP) gave similar and highest grafting levels at all temperatures. However, the GMA-DVB grafting reaction conducted in the presence of DCP resulted in higher amount of poly-GMA formation (albeit formed at much lower concentration than in the conventional system) and more PP branching (lower MFI) compared to a similar system initiated by T101 (not shown), therefore the peroxide T101 was used for all other GMA-DVB samples.

One of the main problems in the GMA-conventional grafting system is the fact that high peroxide concentrations are required to achieve higher grafting levels, but it was shown that this would also give rise to higher extent of both of the main unwanted side reactions: polymer degradation and high poly-GMA formation, see Fig. 1. We have shown previously [13,14] that in the presence of the highly reactive polyfunctional comonomer TRIS, used to assist the

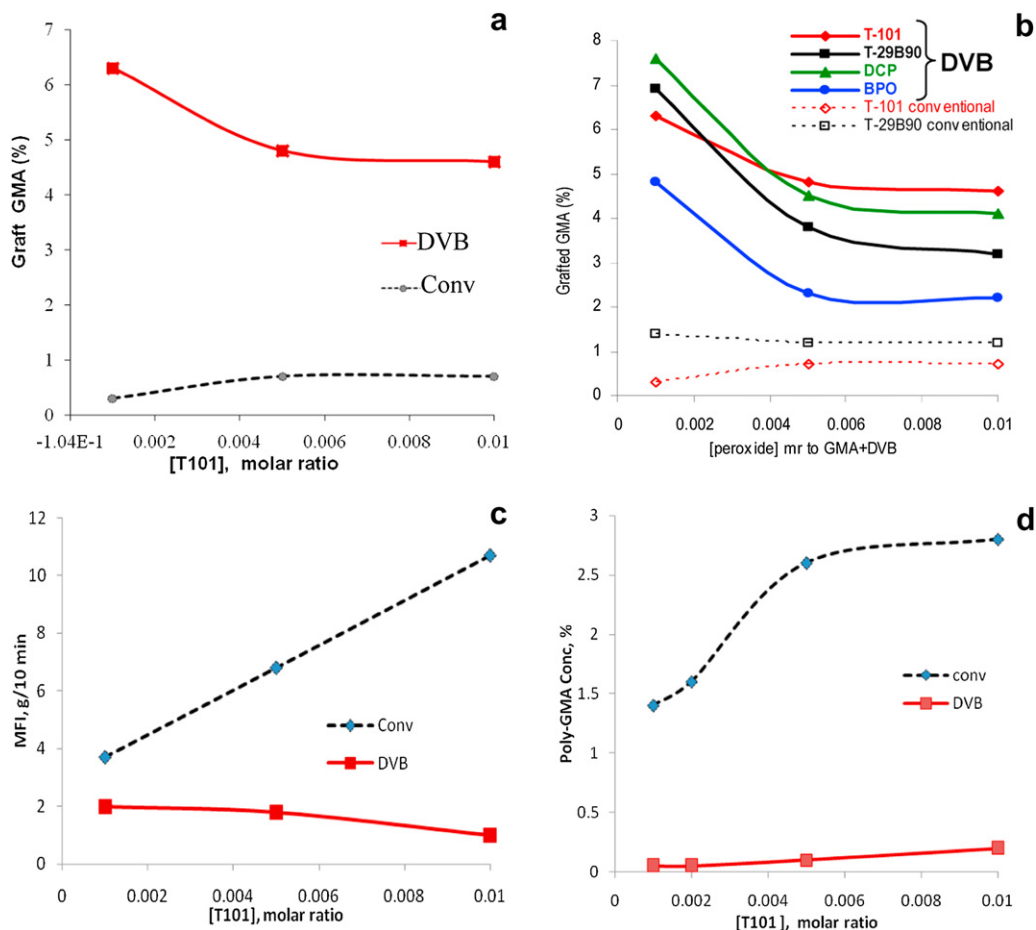


Fig. 7. Effect of concentration of different peroxides (a and b) on GMA grafting level and the effect of the peroxide T101 on changes in MFI of the polymer (c) and the extent of poly-GMA formation (d) in GMA-DVB (at 8:2 w/w) and conventional grafting systems ([GMA] = 12%, $T = 180^\circ\text{C}$). BPO is benzoyl peroxide, DCP is dicumyl peroxide and T-29B90 is Trigonox 292-B90 (1,1-di(*tert*-butylperoxy)-3,3,5-trimethyl cyclohexane). MFI of virgin PP = 2 g/10 min.

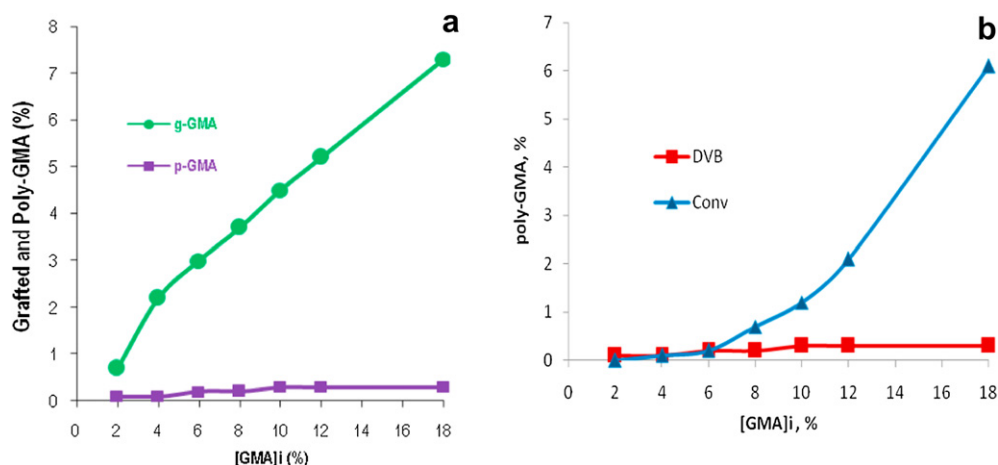


Fig. 8. Effect of GMA initial concentration on the extent of grafting and homopolymerisation of GMA in GMA-DVB and conventional systems ($T = 180^\circ\text{C}$, $[\text{T101}] = 0.01 \text{ mr}$).

grafting of different monomers on polymers, a very small amount of peroxide is needed to achieve high grafting levels. Similarly, one of the main advantages of using DVB in the GMA-grafting system is that it requires only a very small concentration of peroxide to achieve high GMA grafting efficiency. Fig. 7 illustrates this clearly in GMA-DVB grafting reactions initiated by different amounts of different peroxides and shows that the grafting level is highest at the lower peroxide concentrations, for example, with a very low concentration of 0.001 molar ratio, e.g. for ratio of the peroxide T101 to DVB + GMA, a grafting level of over 6% is achieved compared to only 0.3% in a similarly processed conventional GMA (in absence of DVB) grafting system (Fig. 7a and b). The consequence of using such low peroxide concentrations is that very little PP degradation occurs in the GMA-DVB grafting systems, e.g. initiated with T101, as shown from MFI results whereby at such low concentrations almost no change (compared to MFI of the virgin polymer) is observed and at higher peroxide concentrations there is a slight decrease in the MFI, Fig. 7c, reflecting some extent of DVB-assisted branching. This behaviour contrasts sharply with that of the conventional system where higher MFI values are observed and these continue to increase with increasing the initiator concentration reflecting the severity in PP degradation through chain

scission reactions, Fig. 7c. The second most damaging side reaction that competes efficiently with, and occurs at the expense of, grafting in the conventional system is the formation of poly-GMA. The amount of poly-GMA formed during different peroxide (T101)-initiated reactions of GMA-DVB in PP was therefore examined. Fig. 7d shows that, compared to the conventional system, the amount of poly-GMA in the GMA-DVB system decreases to an insignificant level at all peroxide concentrations examined, and this was also the case at all processing temperatures and with different peroxides (results not shown).

The effect of increasing the initial GMA concentration in the GMA-DVB grafting system is to increase the level of grafting but without any significant effect on the extent of Poly-GMA formation which remains to be insignificant, well below 0.3%, Fig. 8. This contrasts the behaviour of a similar system but processed in the absence of DVB (conventional) where the level of poly-GMA increases continuously reaching to a very high level of above 6% at 18% initial GMA concentration, Fig. 8b. This finding that high grafting levels in the GMA-DVB grafting system are paralleled by very low degree of GMA homopolymerisation observed when high initial GMA concentrations are used in the grafting system is an important one as it suggests that the GMA-DVB grafting reaction

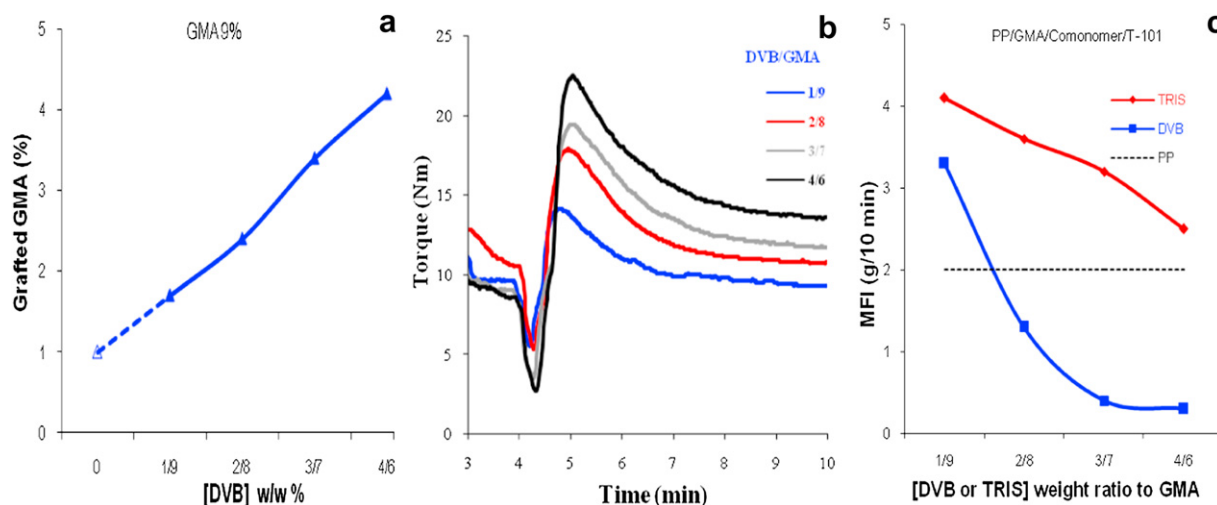


Fig. 9. Effect of amount of DVB used on the extent of GMA grafting (a) the torque behaviour (b) and changes in MFI (c) during processing ($T = 180^\circ\text{C}$, $[\text{T101}] = 0.005 \text{ mr}$, $[\text{GMA}] = 12\%$ except for a where 9% was used).

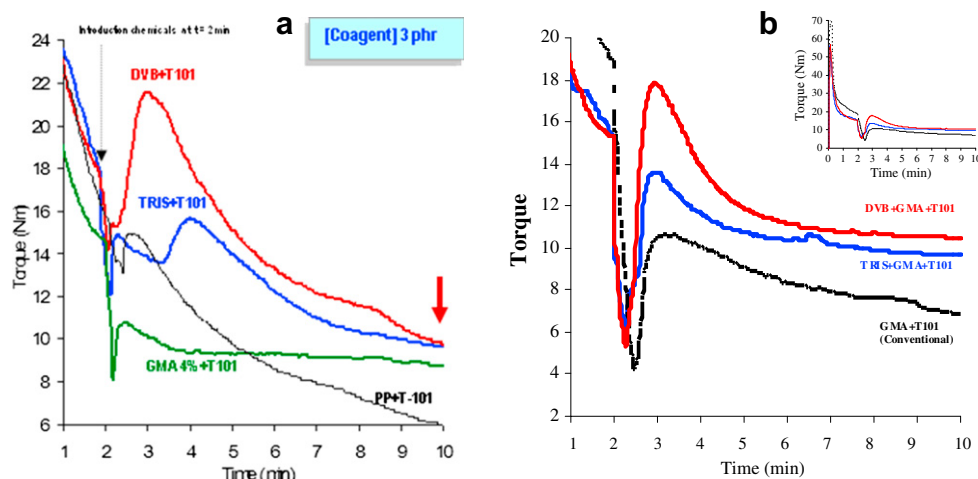


Fig. 10. Torque-time curves for different grafting systems, ratio of GMA to DVB or Tris = 7:3 w/w, [GMA] = 12% except where otherwise indicated on the curve, [T101] = 0.005 mr, $T = 180^{\circ}\text{C}$.

dominates significantly and undermines the otherwise more important GMA–GMA coupling reaction that would otherwise dominate under such high initial GMA concentrations as is the case in the conventional system. This then supports the suggestion that the high reactivity of DVB towards both PP-macroradicals and GMA

would ultimately result in predominance of the DVB reaction with GMA, rather than coupling reaction of GMA–GMA radicals leading therefore to higher extent of grafting associated with almost no poly-GMA formation. Higher concentration of DVB in the grafting system resulted in further increase in the grafting level (Fig. 9a) but

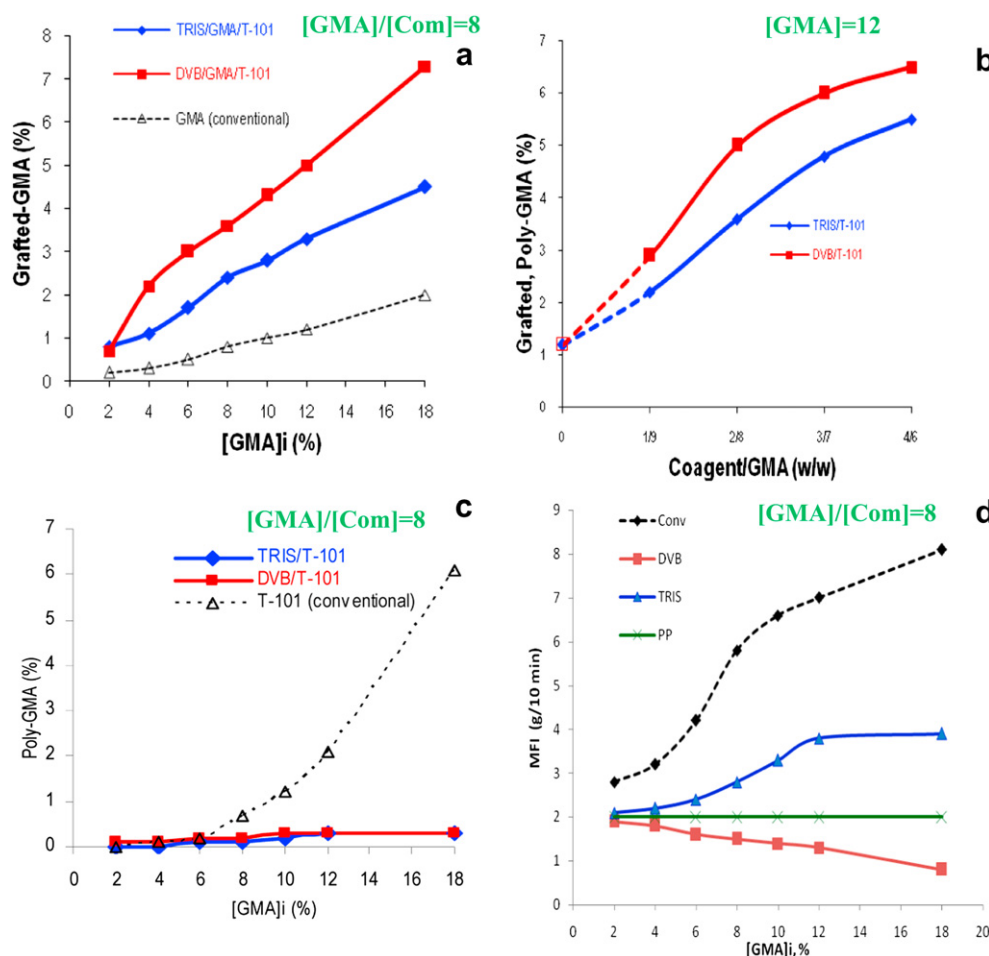


Fig. 11. Effect of initial GMA concentration and ratio of GMA/comonomer (TRIS and DVB) on the level of grafting and homopolymerisation of GMA (a–c) and on the MFI (d) of the polymer ([T101] = 0.005 mr, $T = 180^{\circ}\text{C}$).

this was accompanied by higher extent of branching and cross-linking of the polymer as reflected by an increase in both the height of the torque peak, the final torque values paralleled by a corresponding decrease in MFI (Fig. 9b and c, respectively) due to increase in polymer viscosity.

3.2.2. Comparison of the effect of DVB with that of TRIS on the PP-GMA grafting reaction

The results presented above have clearly shown the remarkable benefits in using the bifunctional comonomer DVB as a comonomer in the grafting reaction of GMA on PP. Compared to similar grafting system but without the comonomer, the presence of DVB was shown to give rise to a significant increase in the GMA grafting efficiency in PP achieved at a very low peroxide concentration resulting in almost complete absence of poly-GMA and a much reduced polymer degradation. The tri-functional comonomer TRIS has also a high free-radical reactivity; the question is therefore how does it differ from DVB in its reactivity towards PP-macroradicals and GMA, hence its effect on the overall- nature of the GMA grafting reaction and the extent of enhancement of the grafting efficiency of GMA in PP.

Fig. 10a shows that the high torque peak maximum observed during the melt reaction of T101-initiated PP-DVB (without GMA) is reduced when DVB is replaced by the same amount of the comonomer TRIS. The higher torque values for DVB indicates that its reaction with the PP-macroradicals occurs to a higher extent at the early stages compared with the reaction of TRIS. The presence of GMA in the DVB- or TRIS- grafting systems causes the torque peak height to decrease but in both cases the peak remained higher than that in the absence of the comonomers (i.e. in GMA-conventional grafting system), with the DVB-GMA torque peak still higher than that of the TRIS-GMA peak, Fig. 10b. These results strongly suggest that the extent of branching caused by the comonomers at the initial part of the reaction (when used without GMA), albeit to a different extent, is undermined in the presence of GMA (i.e. in GMA-comonomer systems) due to the occurrence of further reactions between the comonomers and the GMA that would result ultimately in high levels of GMA grafting at the end of the processing operation. The higher torque peak value in the DVB-GMA system may indicate a higher grafting level achieved here compared to the TRIS-system. This was indeed found to be the case under all conditions tested, for example, Fig. 11 compares the

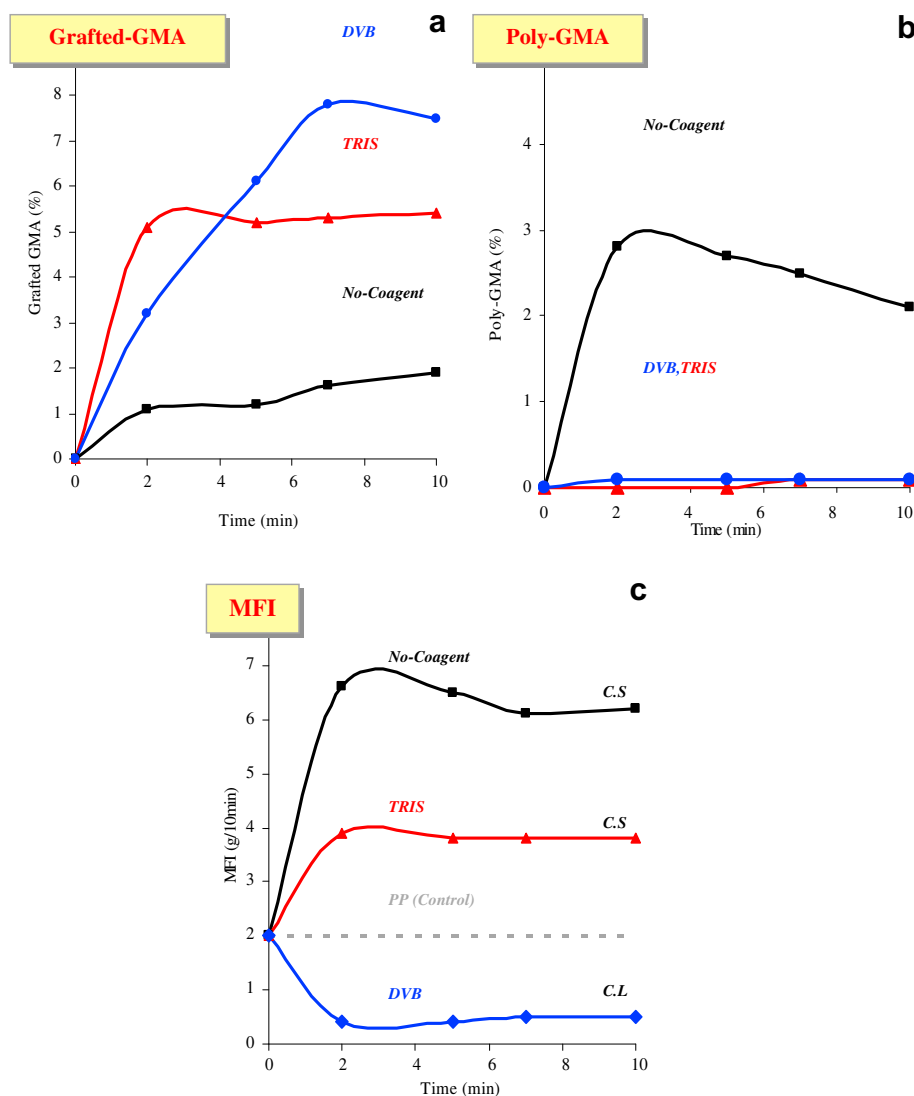


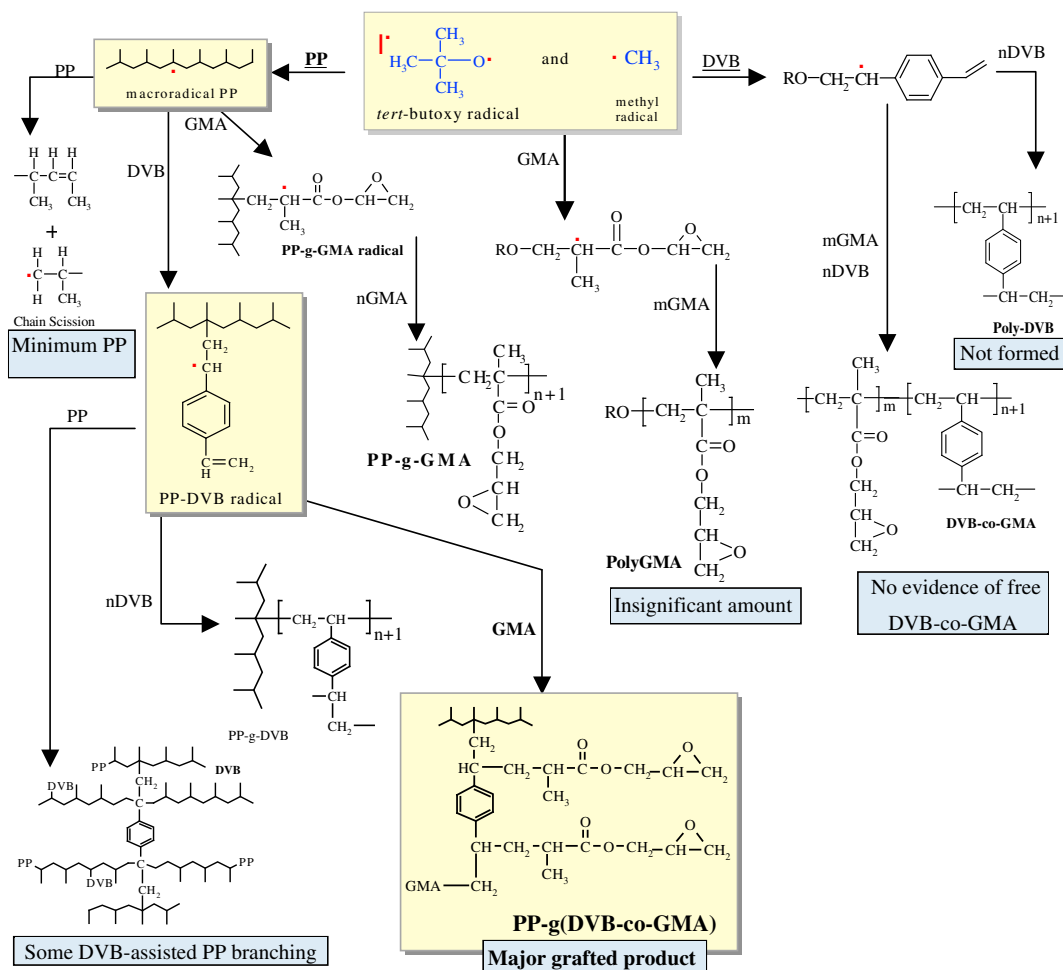
Fig. 12. Rate of GMA graft (a) and homopolymer formation (b) and changes in MFI (c) during melt processing of PP in GMA-Comonomers and conventional systems ([GMA] = 12%, [T101] = 0.005, $T = 160^\circ\text{C}$).

behaviour of DVB with TRIS in the comonomer-GMA grafting systems under different conditions. It is clear that under all conditions examined, e.g. different initial GMA concentrations (Fig. 11a), different GMA-to-comonomers ratios (Fig. 11b), the level of GMA grafting is higher in the presence of DVB (both comonomer systems give much higher grafting level than in the corresponding conventional system). In spite of the difference in the grafting level, both comonomer-containing systems gave rise to an insignificant levels of poly-GMA under all conditions, which contrasts sharply the behaviour of the conventional grafting system where a very high poly-GMA levels can be observed (Fig. 11 c). The effect of DVB and TRIS on the melt flow index of PP-GMA-grafting system is very different. Fig. 11d shows that, compared to the GMA-conventional grafting system, the presence of TRIS results in lowering the extent of PP-chain scission as reflected by the observed lower MFI values (e.g. at all GMA concentrations examined, though still higher than that of the virgin PP), whereas the MFI in the DVB-containing samples is seen to be closer to that of the virgin polymer at lower GMA concentrations but decrease to below the value of PP with increasing GMA concentration. These results suggest that TRIS and DVB are involved in both grafting and branching reactions but that the DVB contribution is higher than that of the TRIS and this is entirely consistent with the higher torque values observed in the DVB-containing system, e.g. see Fig. 10.

The kinetics of the GMA-grafting reactions in the absence and presence of the comonomers was further examined by sampling out at different time intervals during the reactive processing

followed by purification and analysis to determine the nature and yield of the different reaction products and the melt flow behaviour of the polymer during the melt processing. Fig. 12 compares the rate of formation of the grafted GMA (a), poly-GMA (b) and the changes in the melt flow index (c) that take place during the melt reactions in the three systems in the absence (conventional) and presence of DVB and TRIS (all compared under the same conditions). It is clear from this figure that compared to the conventional system (examined under the same conditions), very little poly-GMA is formed in the presence of both comonomers throughout the melt grafting reactions, and this contrasts sharply to the poly-GMA behaviour in the conventional system where its concentration builds up right from the start of the melt reaction resulting in an overall higher final concentration than the amount grafted (Fig. 12b and also Fig. 4).

Further, the rate of grafting in the presence of both comonomers is also, and from the start of the reaction, much higher than that in the conventional system, and the extent of grafting increases further with processing time reaching to a much higher final level (Fig. 12a). However, there is an important and interesting difference in the rate of the grafting reaction when TRIS is present compared to DVB. It is clear from Fig. 12a that in the presence of TRIS, the initial rate of GMA-grafting is faster and is complete within the first 2 min of the reaction, whereas with DVB the initial rate of grafting is much slower reaching to only about 40% in the first 2 min, but it continues to increase with longer reaction times reaching to completion at longer time of about 7 min before levelling off at



Scheme 1. Reaction mechanism of free radical initiated melt grafting of GMA on PP in the presence of DVB.

a much higher grafting level than in the case of the TRIS-system. These results, can be attributed to the differences in the reactivity of the two comonomers towards the macroradicals and the GMA. The three highly reactive acrylate groups in TRIS give rise to the higher initial rate of grafting whereas the two styrenic double bonds in DVB are more resonance stabilised with the aromatic ring, leading to the observed slower but continuous rate of the graft copolymerisation reaction that takes longer time for completion but results in ultimately higher level of grafting.

The lower torque peak value in the peroxide-initiated PP-GMA melt reaction compared to that of PP-DVB or PP-TRIS (with the same peroxide but in absence of GMA) reactions, see Fig. 10a, indicates that the reactivity of GMA towards the PP-macroradicals (resulting in grafting) is lower than the reactivity of DVB or TRIS with the macroradicals. This would then support the suggestion that in the comonomer-GMA-grafting systems the high reactivity of the comonomers towards the macroradicals would result in first grafting of the DVB or TRIS on the polymer backbone before their copolymerisation with GMA to give rise to PP-g- DVB-co-GMA or.

PP-g-TRIS-co-GMA, see Scheme 1 for the main grafting and competing reactions illustrated in GMA-DVB grafting system. The high reactivity of multifunctional comonomers towards macroradicals was also illustrated in other polymer matrices, e.g. TRIS in EPR and EPDM [5,13] and styrene in PP, PE and EPR [19–21].

4. Conclusions

The incorporation of a small concentration of the highly reactive comonomers DVB or TRIS in PP/GMA free radical melt grafting system has been shown to bring about advantageous changes in the grafting yield of GMA. In the presence of a very small concentration of peroxide a faster rate of grafting is achieved concomitant with a substantial reduction in the main competing side reactions, namely chain scission or comonomer-assisted crosslinking of the polymer and the GMA-homopolymerisation (a major competing reaction) where a negligible amount of poly-GMA was obtained under all conditions examined. This behaviour of the GMA-comonomer-containing system contrasts sharply with that of the corresponding conventional system prepared in the absence of a comonomer which required a much higher initiator concentration to achieve a 'good' grafting level which was still much lower than that obtained in

presence of one of the above comonomers, as well as giving rise to much higher levels of poly-GMA and increased degree of polymer degradation.

A comparison of the effect of the comonomers DVB and TRIS on the behaviour of the GMA-grafting system revealed that DVB always gave rise to higher extent of GMA grafting compared to TRIS, and the rate of the grafting reaction was also different whereby the presence of TRIS gave rise to complete grafting reaction in a much shorter time, within the first 2 min of processing, whereas the a much lower initial level of grafting was achieved in presence of DVB but the grafting reaction continued with processing time levelling off at longer time (about 7 min). This difference is a direct consequence of the different reactivity of the two comonomers towards the macroradical and the GMA. Overall, however, both DVB and TRIS act as promoters for the grafting reaction of GMA via an in-situ co-grafting process resulting in a polymer containing highly grafted GMA with low level of degradation and almost without GMA-homopolymer formation.

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